Geopolitical Event-Driven Portfolio Risk Analysis Platform

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GitHub Repository Link: https://github.com/jjeong4335/gdelt-risk-platform

***Abstract*—Traditional market volatility metrics, such as VIX, indicate market fear but fail to isolate causality or quantify portfolio-specific vulnerabilities. To address this for institutional investors, corporate clients and people's personal portfolios, we present a robust analytical platform that models geopolitical risk through an automated, real-time data pipeline. By processing a decade of global news via the GDELT database alongside S&P 500 price data, our system computes a proprietary Geo-Tension Index. Utilizing distributed computing frameworks like PySpark on Google Cloud Dataproc, the platform handles massive datasets seamlessly. This solution detects geopolitical tension spikes and evaluates historical sector-level reactions over a 30-day window. This end-to-end product provides actionable, real-time risk exposure analysis, updating continuously every 15 minutes to explain these vulnerabilities and empower rapid, data-driven financial decision-making.**

***Index Terms*—Geopolitical Risk, Portfolio Analytics, Big Data, PySpark, Sentiment Analysis, Dataproc**

I. Introduction

Global financial markets are profoundly influenced by geopolitical events, yet standard metrics like VIX only reflect aggregate market anxiety without offering actionable causality. When a war breaks out, sanctions are imposed, or a coup happens, markets react - but the nature of that reaction varies significantly by sector, magnitude, and duration. Our enterprise-grade platform bridges this gap by determining exactly why the market is reacting and how a specific portfolio is exposed to that specific cause.

News-flow data like the Global Database of Events, Language, and Tone (GDELT) reveals geopolitical stress before the market reflects these changes. By parsing continuously updated GDELT data and S&P 500 data, using GDELT's precomputed negative sentiment scores and aggregating said data, we construct a PySpark ETL pipeline that helps compute a Geo-Tension Index and help detect events of high tension in the world.

This data can be useful for anyone that maintains a stock portfolio, whether it be a risk manager or a person with their own personal portfolio. By seeing this Geo-Tension Index and what sectors are most vulnerable, one can actively hedge, rebalance or communicate an informed risk before broader market contagion occurs.

II. System Architecture & Methodology

The platform is built around a two-speed data pipeline as seen in Figure 1. On one side, a one-time batch job downloads a decade of GDELT news archives, filters down the data, and feeds that into a PySpark cluster on Google Cloud where three sequential jobs compute a daily Geo-Tension Index. Each day's tension score is compared against the mean and yearly mean and standard deviation per year, and any day that exceeds yearly mean + 3σ is flagged as a spike. We then join each anomaly against pre-fetched S&P 500 price history to measure how each market sector moved in the ±30 days around every event which are written to Parquet files on HDFS. On the other side, the dashboard directly polls GDELT's lastupdate.txt endpoint every 15 minutes and fetches up the latest GDELT GKG file and displays current geopolitical headlines in real time. The dashboard that's visible in the final product sits on top of both layers simultaneously; reading the Parquet files for historical lookups and polling GDELT's lastupdate.txt for the real-time view. The result is a view such that a user sees both a decade of context and what is happening right now in the same interface.

III. Implementation

A. Batch Pipeline

The batch pipeline is implemented in PySpark 3.5 and executed on NYU's Google Cloud Dataproc cluster with YARN. Raw GDELT archives are downloaded and filtered at ingestion time via a shell script using geopolitical keywords — WAR, SANCTION, MILITARY, TERROR, PROTEST, WEAPON, NUCLEAR — simultaneously with download, reducing approximately 2TB of raw data to a 5GB geopolitical subset before any Spark processing occurs. Four discrete `spark-submit` jobs handle index construction, spike detec-tion, sector reaction windowing, and news extraction respec-tively, with each job reading from and writing to HDFS as Parquet to enable re-execution without reprocessing upstream stages.

Batches were chosen over streaming throughout intentionally. Both GDELT and market APIs update in discrete intervals, and real-time streaming would add architectural complexity without adding signal. The 15-minute batch collection interval matches GDELT's own update frequency, making it the appropriate engineering decision.

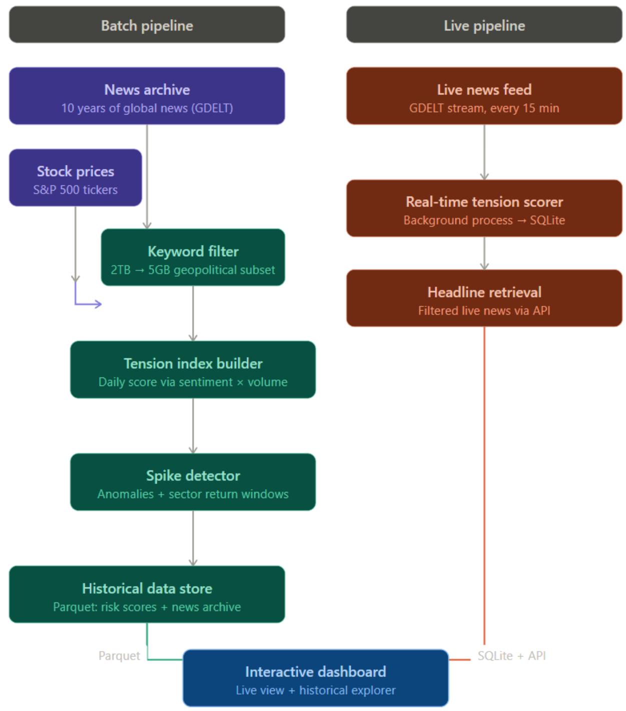

Fig. 1. End-to-end data pipeline for real-time sentiment analysis and historical sector impact analysis .

B. Geo-Tension Index

The daily Geo-Tension Index is formulated as

$$\text{GTI} = \text{avg}(\text{negativetone}) \times \log(\text{articlecount} + 1) \quad (1)$$

, capturing intensity and volume of geopolitical coverage. Raw scores are then normalized to a 0-10 scale using p1-p99 percentile clipping rather than min-max normalization, making the index more robust to outlier distortion. Anomalous spikes are then identified as any day exceeding yearly mean + 3σ of the per-year distribution.

C. Stock Data & Event Windows

Anomalous spikes are identified using a per-year Z-score threshold: any day where the Geo-Tension Index exceeds yearlymean + 3σ is flagged as a spike event. The threshold is computed per year rather than globally, since the Russia-Ukraine War era generated substantially more alarming coverage than 2018 or 2019 — a global threshold would bias detection toward recent high-volume years and miss earlier events. We tested thresholds of 1.5σ, 2σ, and 3σ; lower thresholds captured too much routine noise, while 3σ produced the cleanest separation between everyday news cycles and genuine geopolitical shocks.

D. Live Pipeline

The live pipeline is implemented directly within the dashboard (app. py), which polls via `GDELT's lastupdate.txt endpoint`every 15 minutes to fetch the latest GDELT GKG file and display current geopolitical headlines in real time. A timestamped row is then appended to a local SQLite database. Headline retrieval for the dashboard's news panel fetches directly from GDELT's lastupdate.txt, with a multi-layer domain-based filter applied to prioritize established outlets and suppress low-quality aggregators.

E. Dashboard

The dashboard is built with Streamlit and Plotly, served locally on port 8501 and exposed for demonstration via ngrok. It consumes both pipeline layers simultaneously: Parquet for historical lookups and GDELT's lastupdate.txt for real-time scoring. This results in a unified view of geopolitical tension and market reaction.

IV. Analysis

Predictive Sector Impact: By aggregating historical data across major geopolitical events, the system provides expected sector movements during periods of elevated tension. En-ergy and Defense consistently outperform during geopolitical shocks, while Technology and Airlines tend to underperform. Gold behaves as a safe haven during high-uncertainty events. Critically, the magnitude of these reactions is consistent across events — not one-off cases — which is what transforms an observation into a replicable signal.

Transparent Risk Attribution: The platform directly surfaces the underlying catalyst behind any risk score rather than producing opaque outputs. For example, during similar Middle East tensions, the system can surface that 23 related articles reported an average −11.8% drop in energy stocks, grounding the score in verifiable historical evidence.

3. Portfolio Optimization: Portfolio managers can visualize sector-level exposure during unfolding events and strategically hedge or reallocate assets before broader market contagion oc-curs. Because the Geo-Tension Index leads VIX, the platform provides an earlier signal than waiting for implied volatility to spike.

V. Discussion

The deployment of this platform validates the hypothesis that quantified geopolitical sentiment can serve as a predictive indicator for sector-specific volatility. While traditional risk models rely heavily on historical price action, our pipeline introduces a crucial semantic dimension to risk assessment.

A key architectural strength is the use of Google Cloud Dataproc and PySpark to manage the computational load of cross-referencing a multi-terabyte unstructured news dataset with a decade of financial time-series data. The intentional separation of batch and live pipelines keeps the system maintainable and avoids the complexity of full streaming while still delivering 15-minute update granularity that matches GDELT's own cadence.

Limitations include the reliance on GDELT's machine-generated tone scores, which are efficient at scale but present trade-offs in capturing nuanced diplomatic rhetoric seen in the news. Ticker selection for sector representation reflects manual curation, and stock price data retrieved via `yfinance` was not explicitly verified for split and dividend adjustment consistency across the full decade.

VI. Conclusion & Future Work

We have delivered a scalable Big Data platform that transforms abstract geopolitical sentiment into tangible, predictive financial intelligence. It reveals not how markets move, but why they move, and doing so entirely through scalable data engineering rather than black-box modeling. By integrating distributed computing with automated sentiment aggregation and a dual-speed pipeline architecture, the platform provides investors and risk managers with a critical edge in navigating an increasingly volatile global landscape.

Looking ahead, strategic product enhancements will focus on three core areas:

- **Advanced LLM Integration:** Transitioning from GDELT's pre-computed sentiment scores to fine-tuned Large Language Models to capture deeper contextual nuances.
- **Expanded Data Infrastructure:** Scaling the existing big data architecture to possibly ingest higher velocity alternative data streams.
- **Full-Stack Customization:** Evolving the Streamlit presentation layer into a comprehensive web application that supports granular, user-defined portfolio structures, multi-tenant enterprise access, and automated risk alert systems.